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Project Documentation

Project Overview

Purpose

This application provides a centralized platform for bird-watching enthusiasts to discover and purchase bird-themed products, offering a curated selection of merchandise such as mugs, ties, wine bottles, and other unique items designed for bird lovers and collectors.

Technology Stack

- Frontend: productList.html,
orders.html,
checkout.html
- Backend: jsfile.js
server.js
- Database: SQLite
- Hosting: Local Development Environment

Product Features

Feature 1: productList.html as the main page

Description

Provides an overview of user activity.

Capabilities

- View items
- add the items to cart

Feature 2: Data Management

Description

Supports CRUD operations and transactional data handling for user purchases.

Capabilities

- Create records (e.g., add items to cart / create cart entries)
- Read records (e.g., view cart contents, order summary)
- Update records (e.g., change item quantity in cart)
- Manage transactional workflows (add to cart ==> checkout ==> order creation)

Feature 3: Backend Server and Deployment Setup

Description

Implements a local server environment to run the ecommerce application and handle communication between the frontend and backend.

Capabilities

- Runs application using Node.js and Express
- Serves frontend and backend through a local server
- Handles API requests (e.g., cart operations)
- Connects application to SQLite database
- Enables proper deployment workflow (localhost testing)

Challenges

Challenge 1: CODIO Issues

Problem

The application failed to add stuff to cart during deployment in module six in CODIO, but worked quite fine on local host.

I tried to click the add-to-cart with everything uploaded to Codio. However, it says on the popped-up webpage that Cart data is missing or invalid. In the meantime, in fact, I have uploaded everything to Codio without a public folder.

Solution

The instructor helped debug CODIO. Since the next module, I started to run the application on local host too.

Outcome

Successfully deployed the application on local host.

Challenge 2: CurrentUser is perfect for the execution policy.

Even though it's my personal laptop, Windows installs Node.js with system-level scripts that require administrator-level PowerShell to modify the LocalMachine policy. This is a Windows security design. It protects system-wide settings even from the laptop's own user account unless I choose to explicitly "Run as Administrator." The CurrentUser scope only changes the policy for my own user session, so it doesn't need that elevated permission. I need to think of it this way: LocalMachine affects everyone who uses that computer, so Windows requires a higher level of trust to change it. CurrentUser only affects myself, so no admin elevation is needed. For this coursework, CurrentUser is perfectly fine.

Problem

I received a red error saying that I need admin rights.

Solution

I finally used the alternative (no admin rights needed).

Outcome

For my coursework, CurrentUser is perfectly fine.

Configuration

Development Environment

| Component |

|-----|

| Node.js |

| Express |

| SQLite |

| HTML/CSS/JavaScript |

Installation

1. Clone the repository

This application repo is divided into several separated git repositories.

frontend:

```
```bash
```

```
git clone https://github.com/labricot-lab/m5_lab5_20260310.git
```

```
```
```

```
```bash
```

```
git clone https://github.com/labricot-lab/446_m5_coursepro_20260312.git
```

```
```
```

backend:

```
```bash
```

```
git clone https://github.com/labricot-lab/446_pro_08_20260421.git
```

```
```
```

```
```bash
```

```
git clone https://github.com/labricot-lab/446_pro_07_20260419.git
```

```
```
```

```
```bash
```

```
git clone https://github.com/labricot-lab/446_pro8_updated_20260506.git
```

...

Documentation:

The GitHub documentation link will be generated during the process of uploading this final

application documentation to GitHub.

[https://github.com/labricot-lab/446final\\_20260611](https://github.com/labricot-lab/446final_20260611)

live link:

[https://labricot-lab.github.io/446final\\_20260611/](https://labricot-lab.github.io/446final_20260611/)

2. Install dependencies

```
```bash
```

```
npm install
```

```
```
```

4. Start the application

```
```bash
```

```
node server.js
```

```
```
```



---

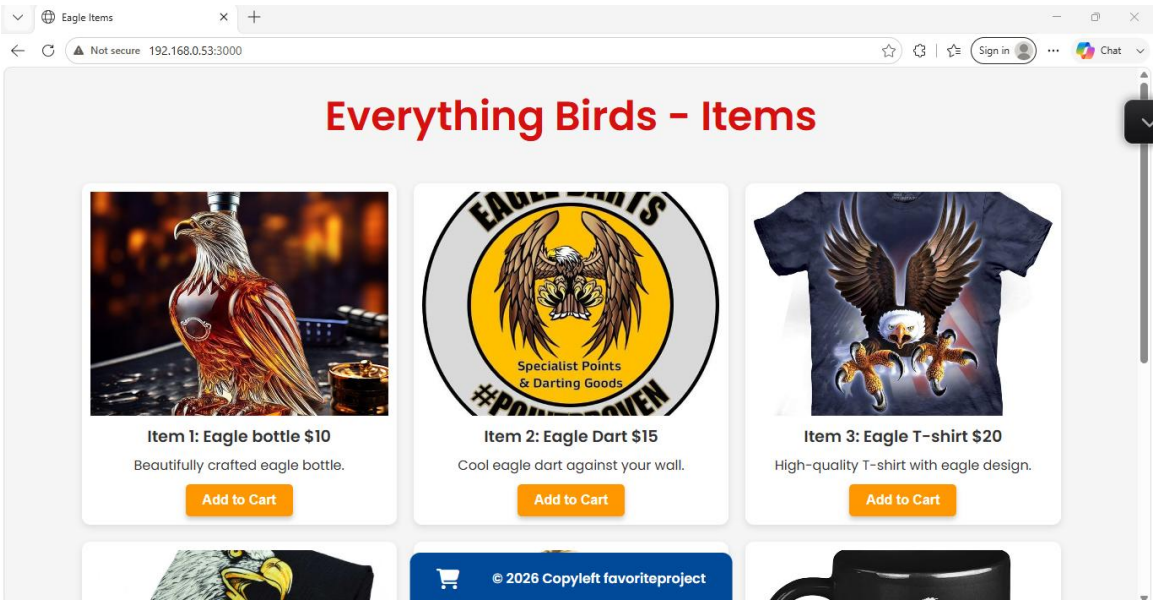
## Testing

| Test Case                  | Status |
|----------------------------|--------|
| -----                      | -----  |
| View Product List          | Passed |
| Add Item to Cart           | Passed |
| Update Cart Quantity       | Passed |
| Checkout Process           | Passed |
| Order Creation             | Passed |
| SQLite Database Operations | Passed |

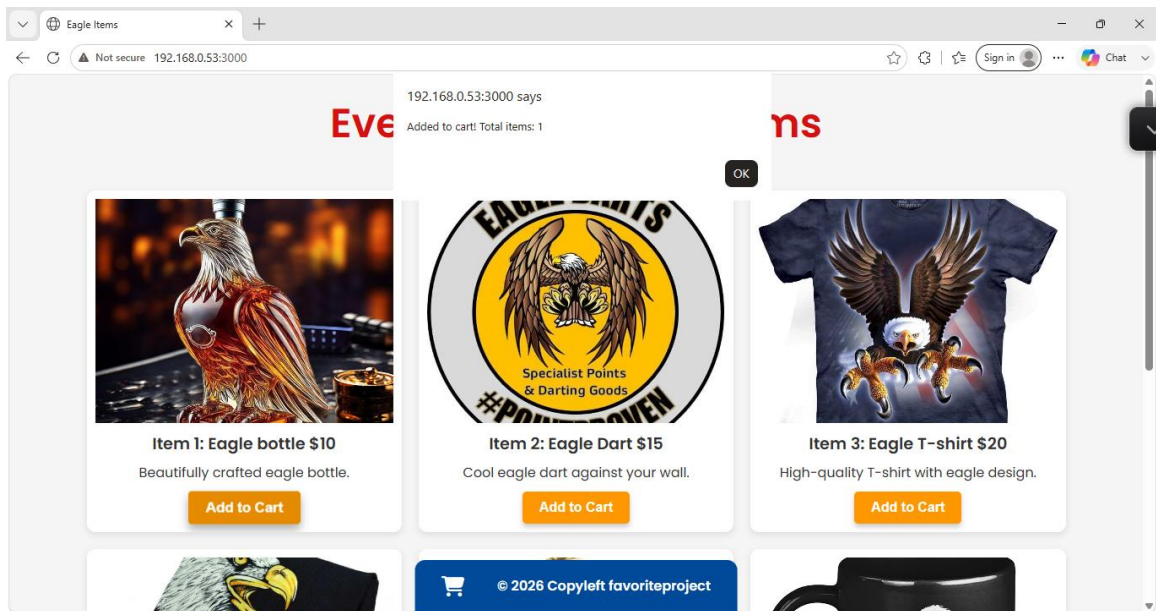
### Testing this application on different browsers and iPhone.

#### Testing on Edge:

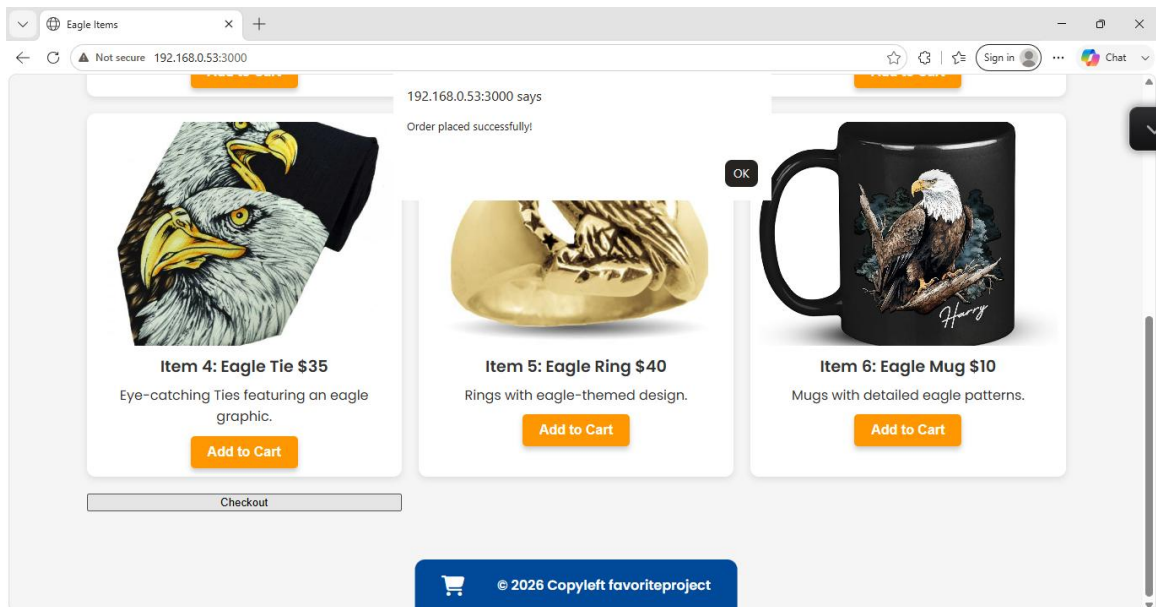
1



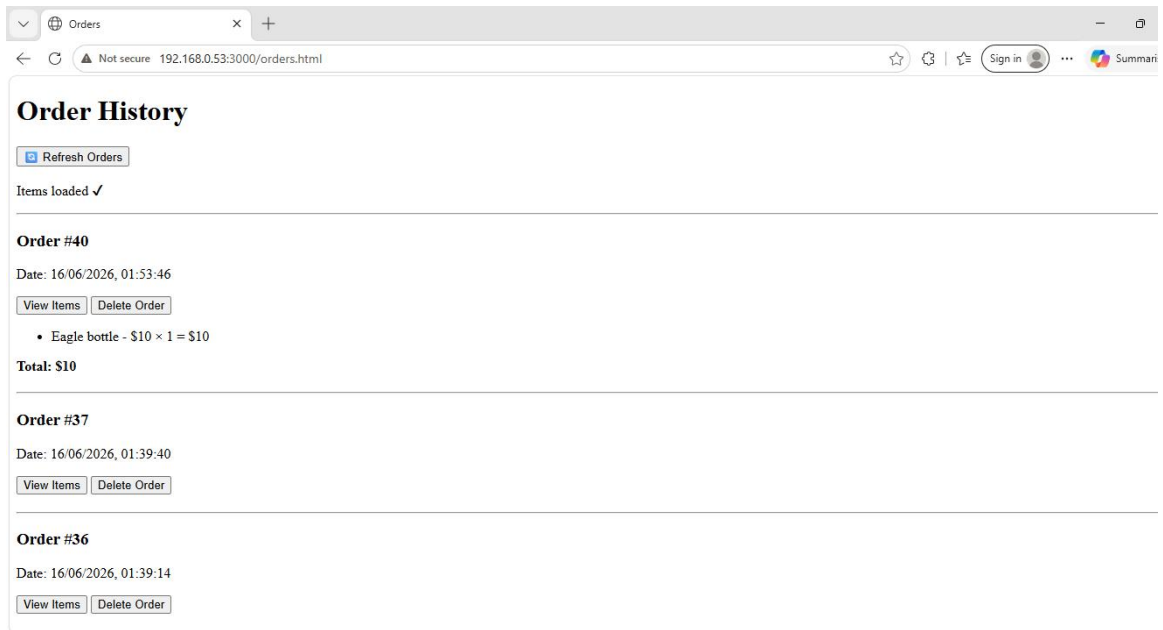
2



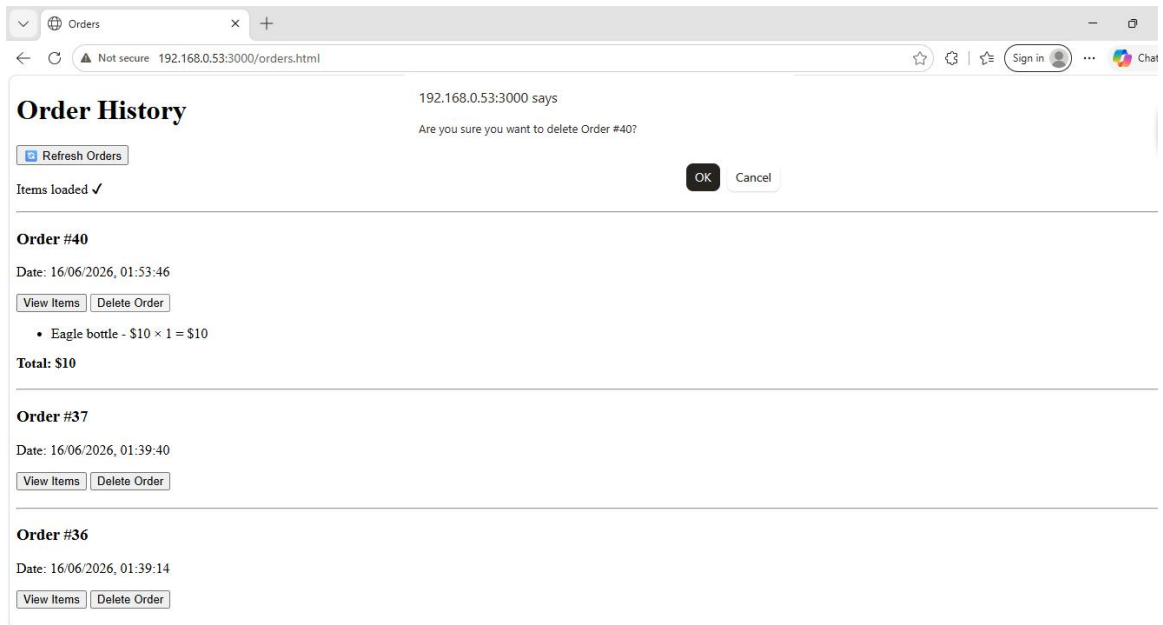
3



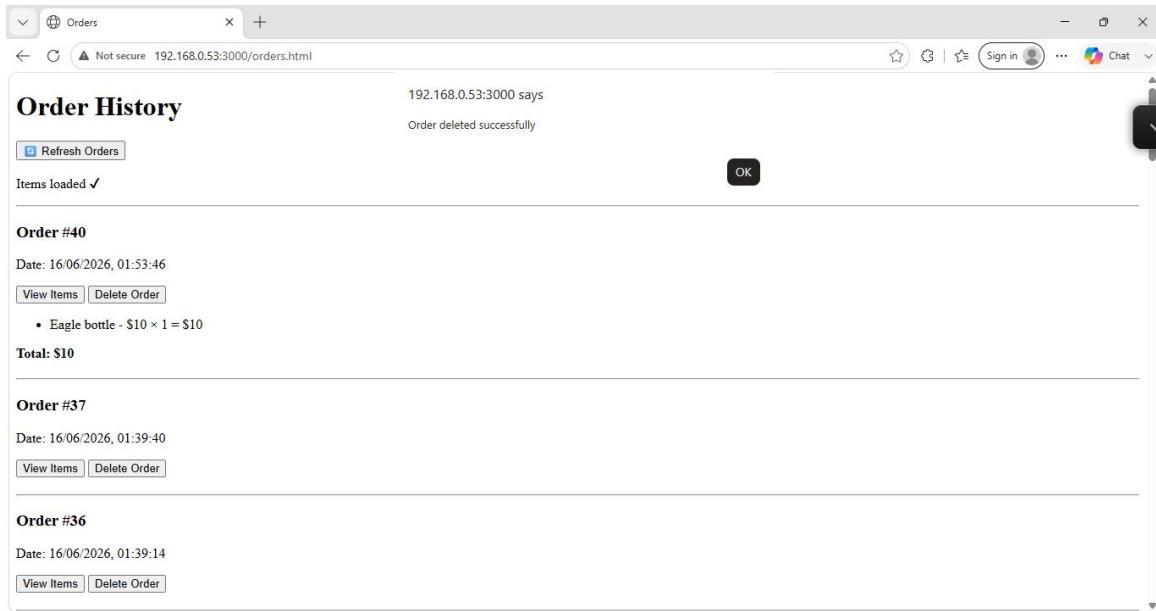
4



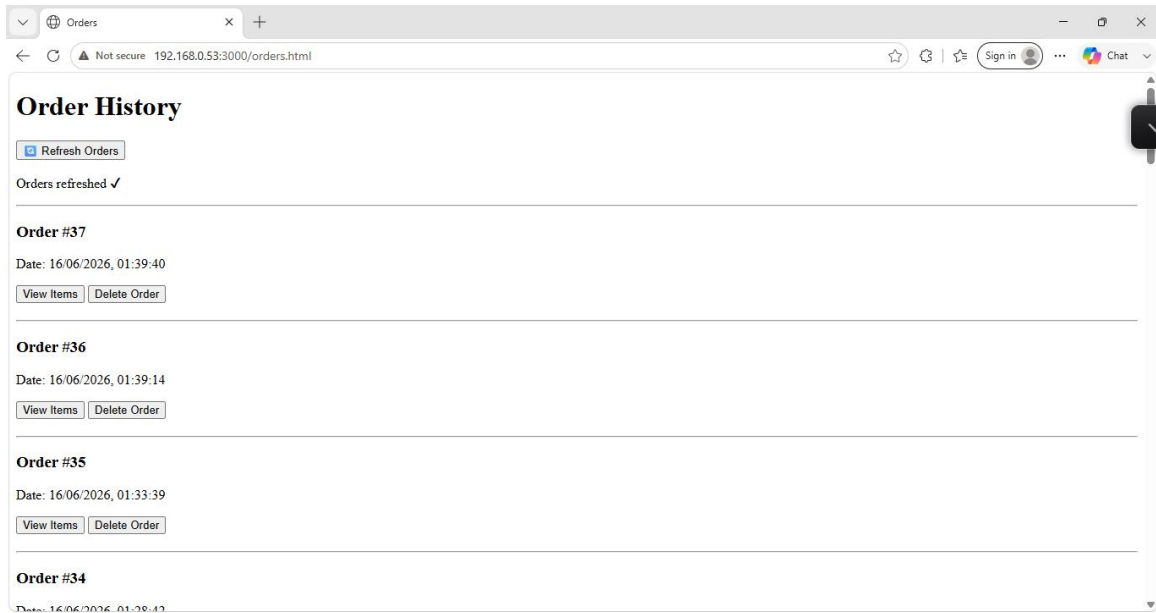
5



6

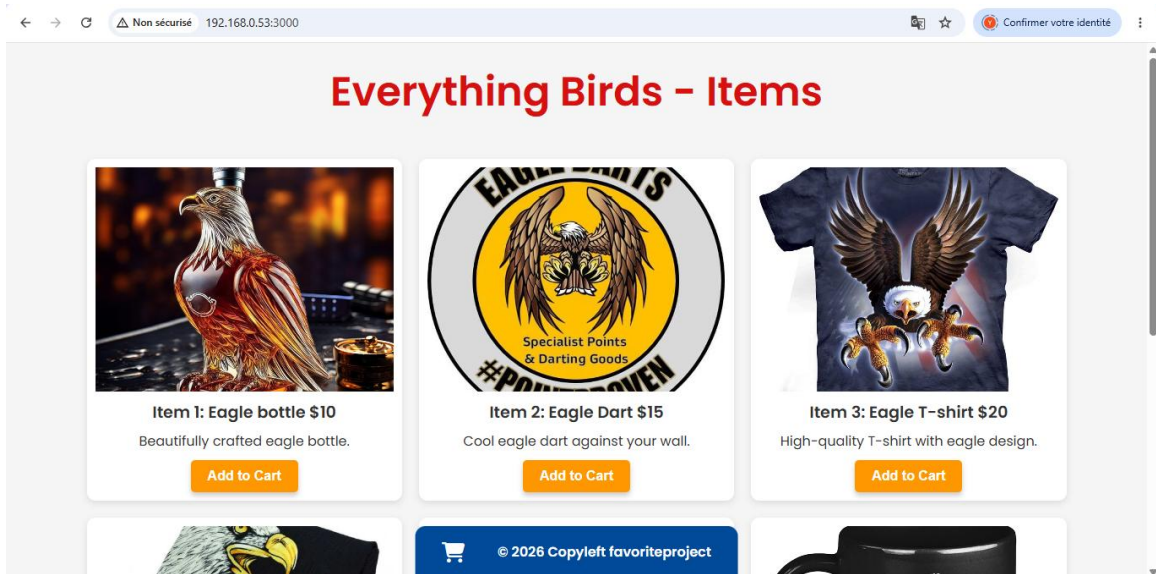


7

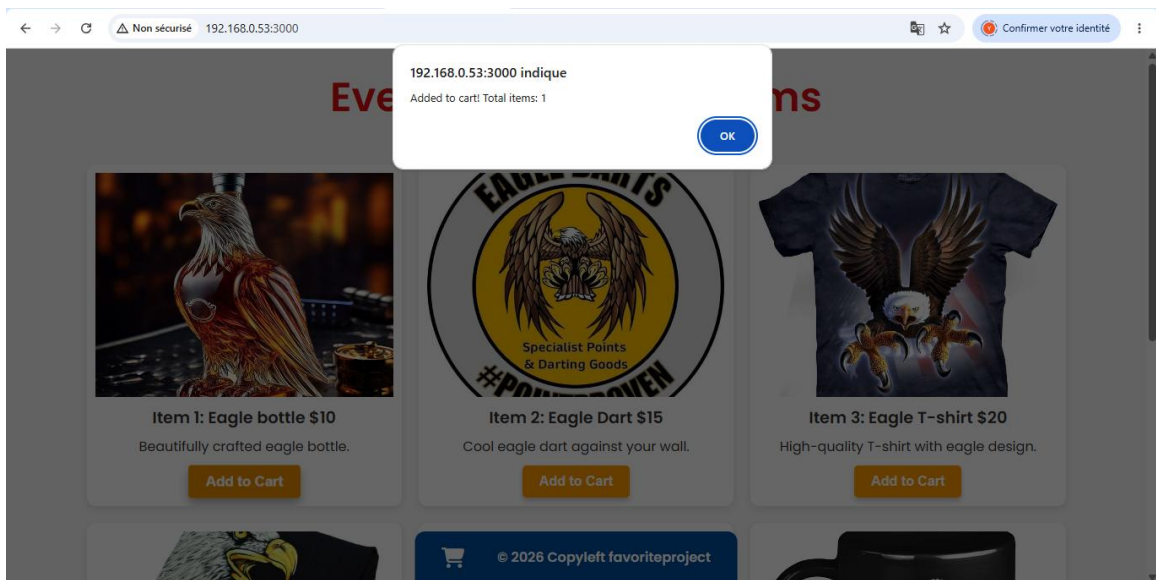


#### Testing on Chrome browser

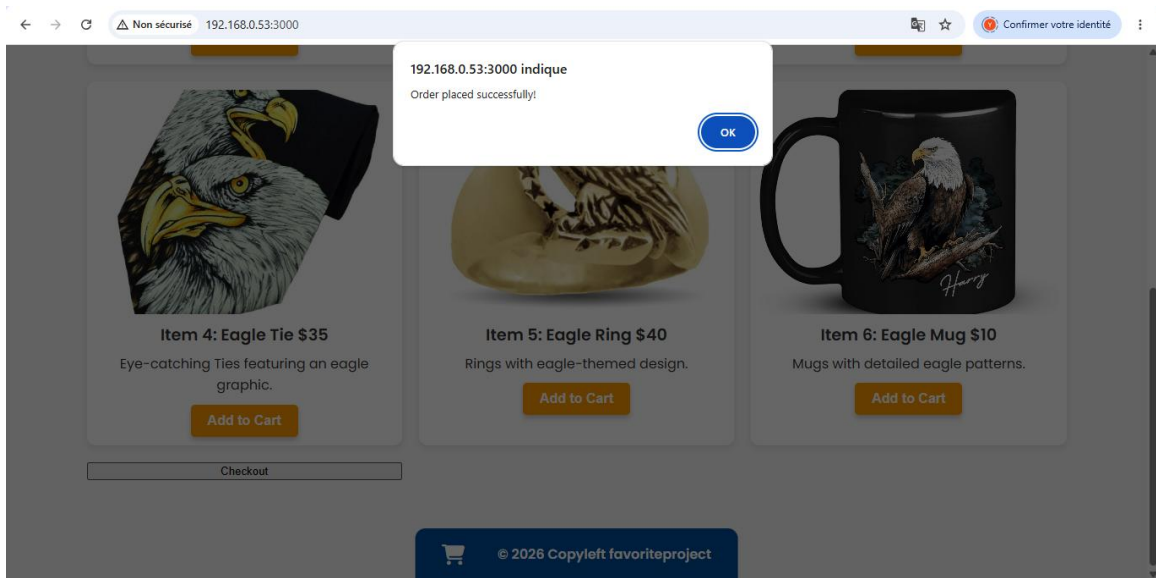
1



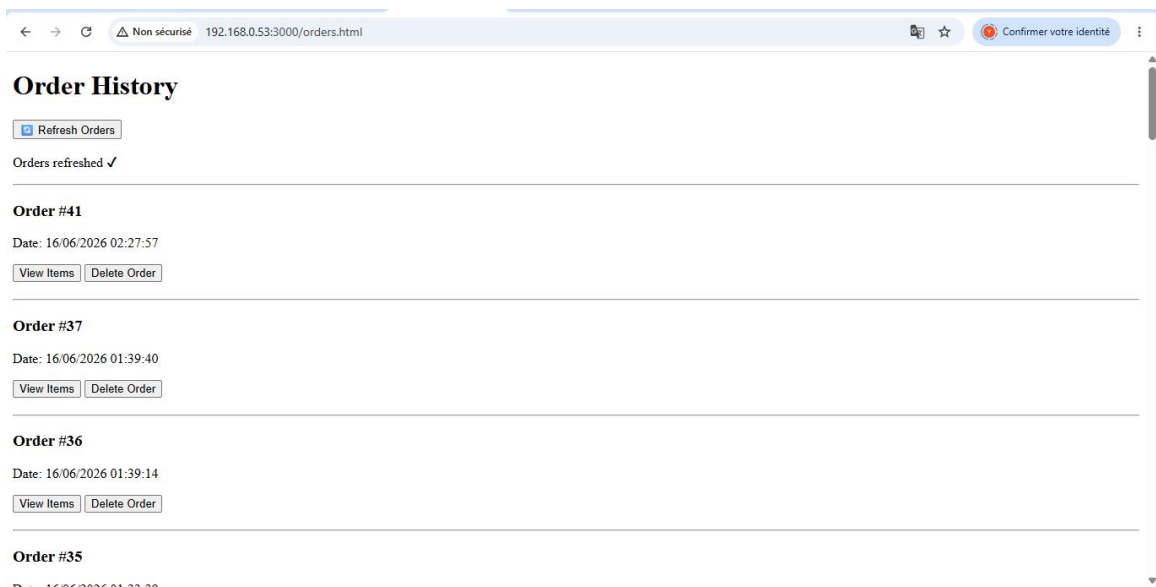
2



3



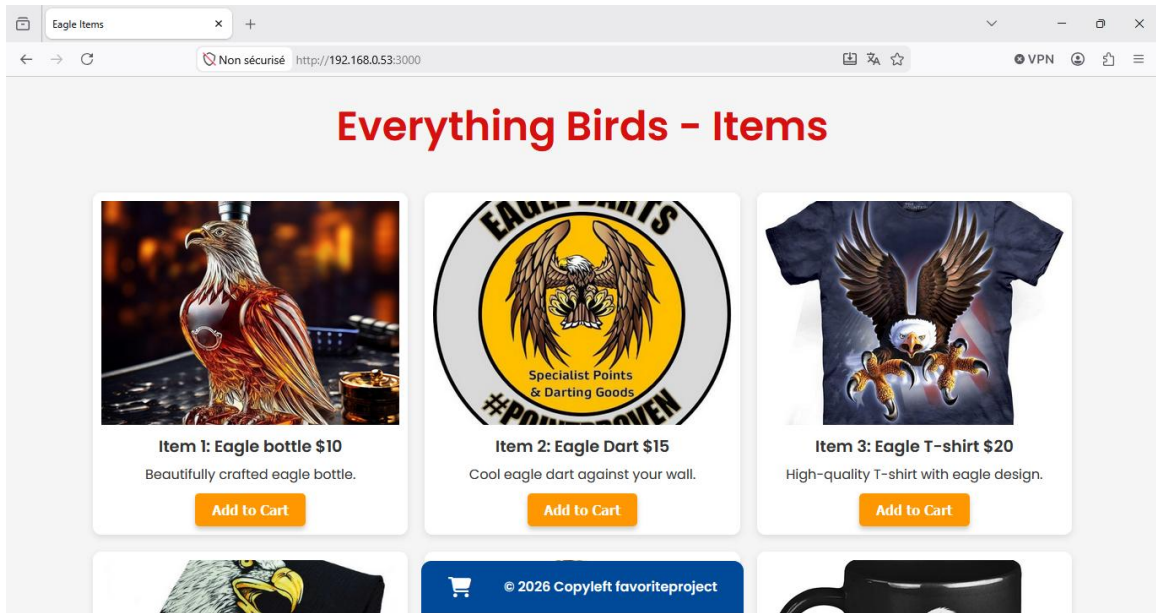
4



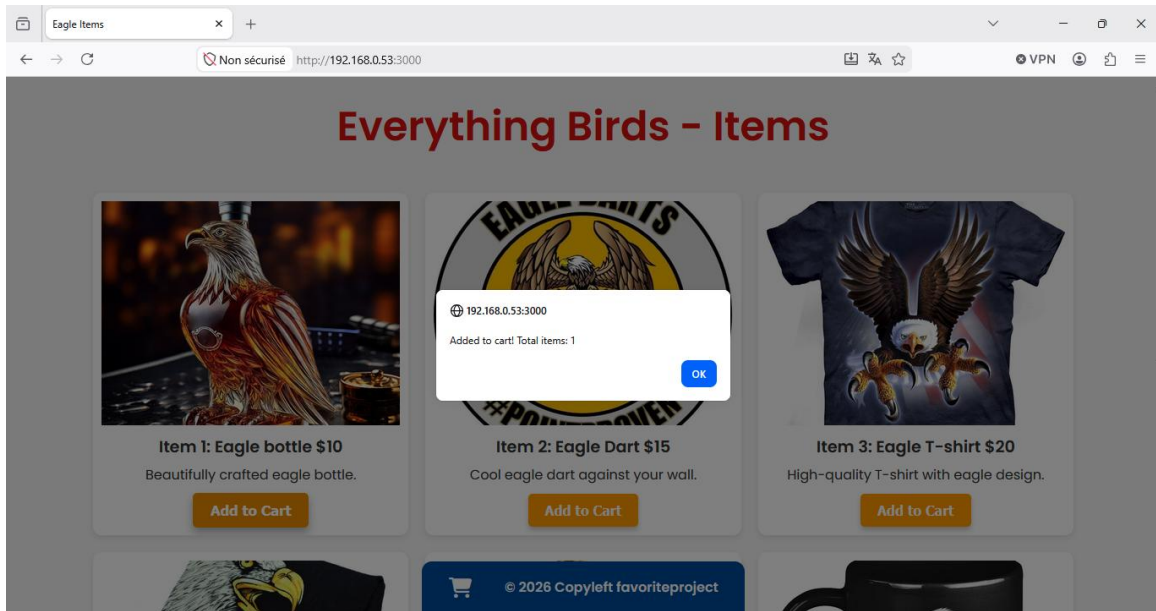
#### Testing on Firefox browser:

1

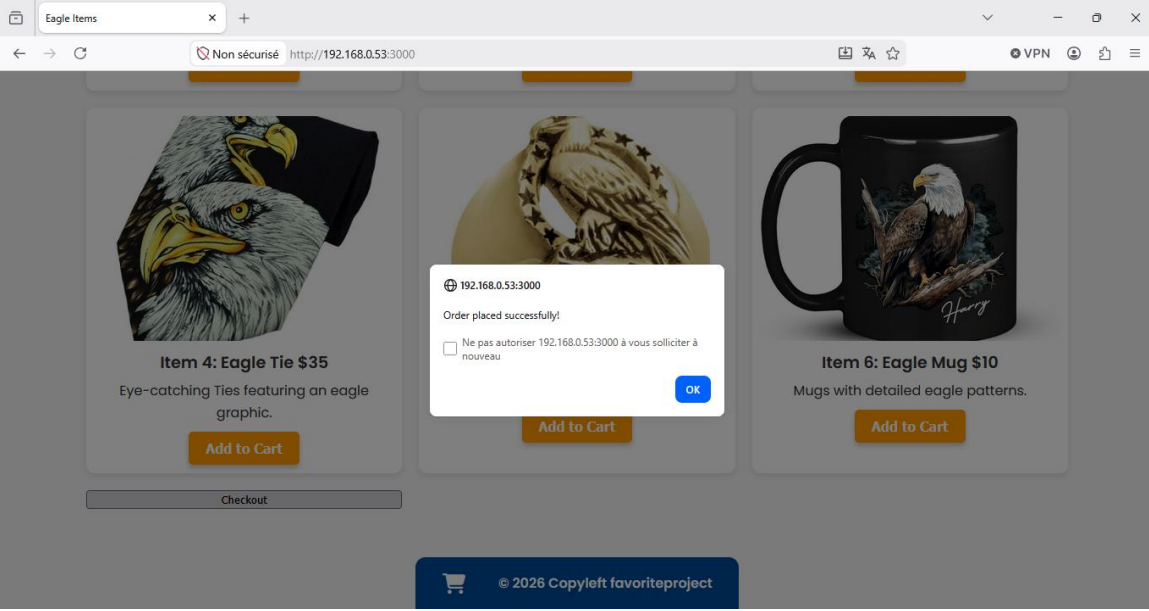




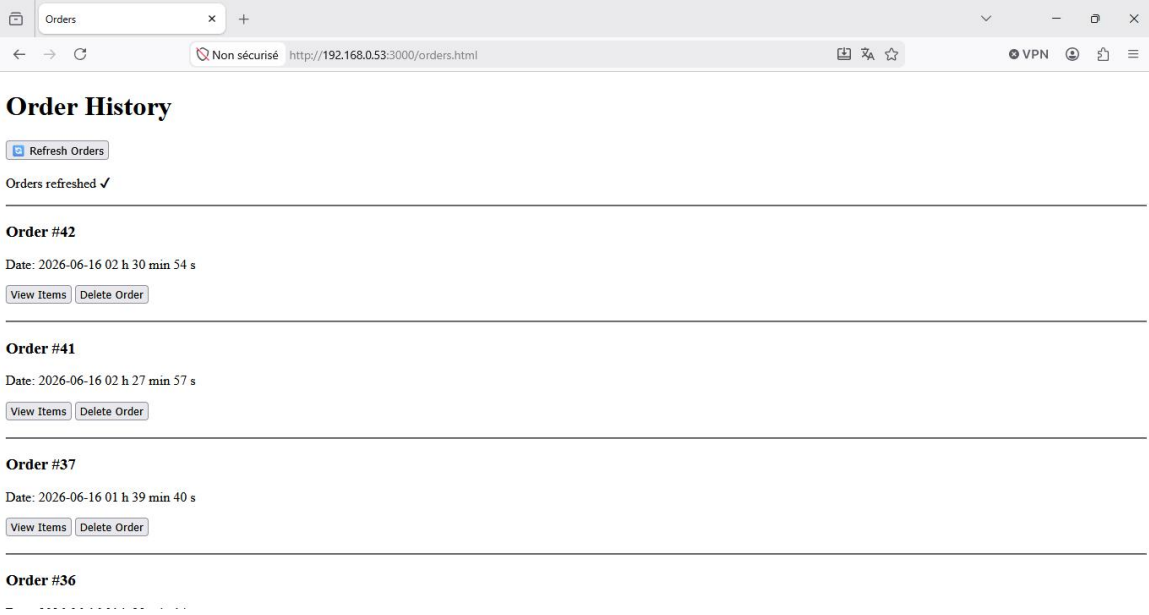
2



3

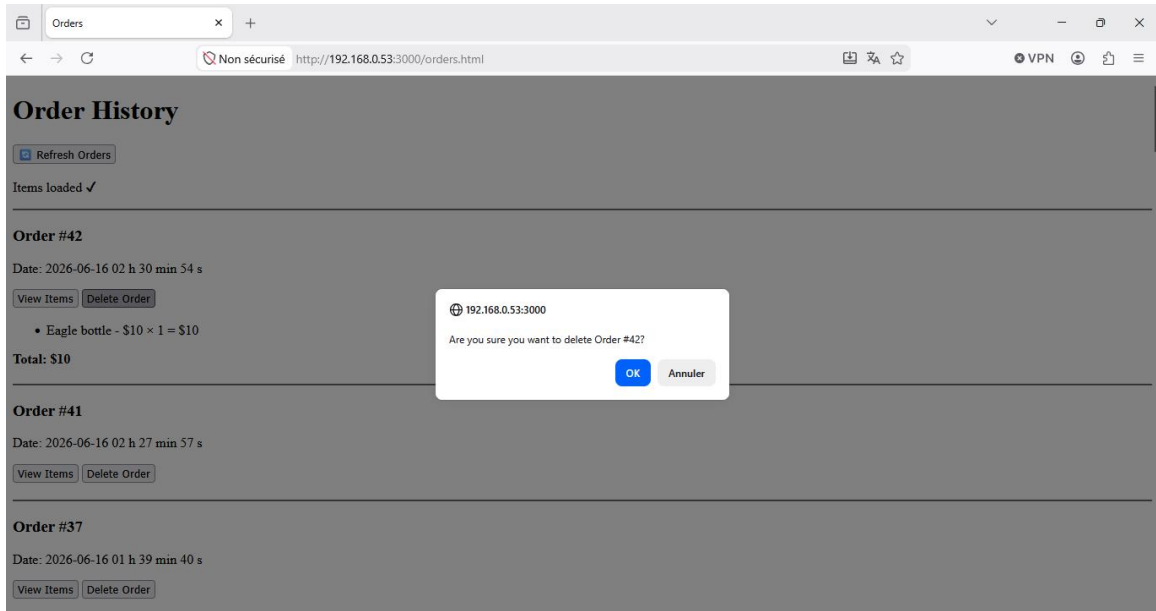


4

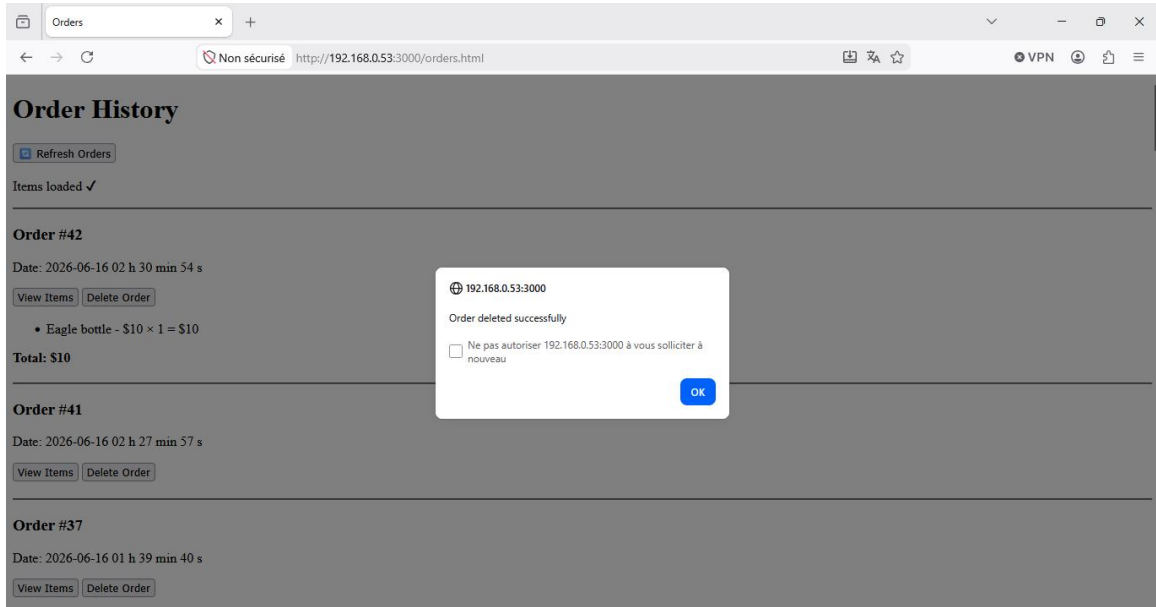


5





6



7

# Order History

Refresh Orders

Orders refreshed ✓

## Order #41

Date: 2026-06-16 02 h 27 min 57 s

View Items

Delete Order

## Order #37

Date: 2026-06-16 01 h 39 min 40 s

View Items

Delete Order

## Order #36

Date: 2026-06-16 01 h 39 min 14 s

View Items

Delete Order

## Order #35

#### Testing on iPhone

# Everything Birds – Items



## Item 1: Eagle bottle \$10

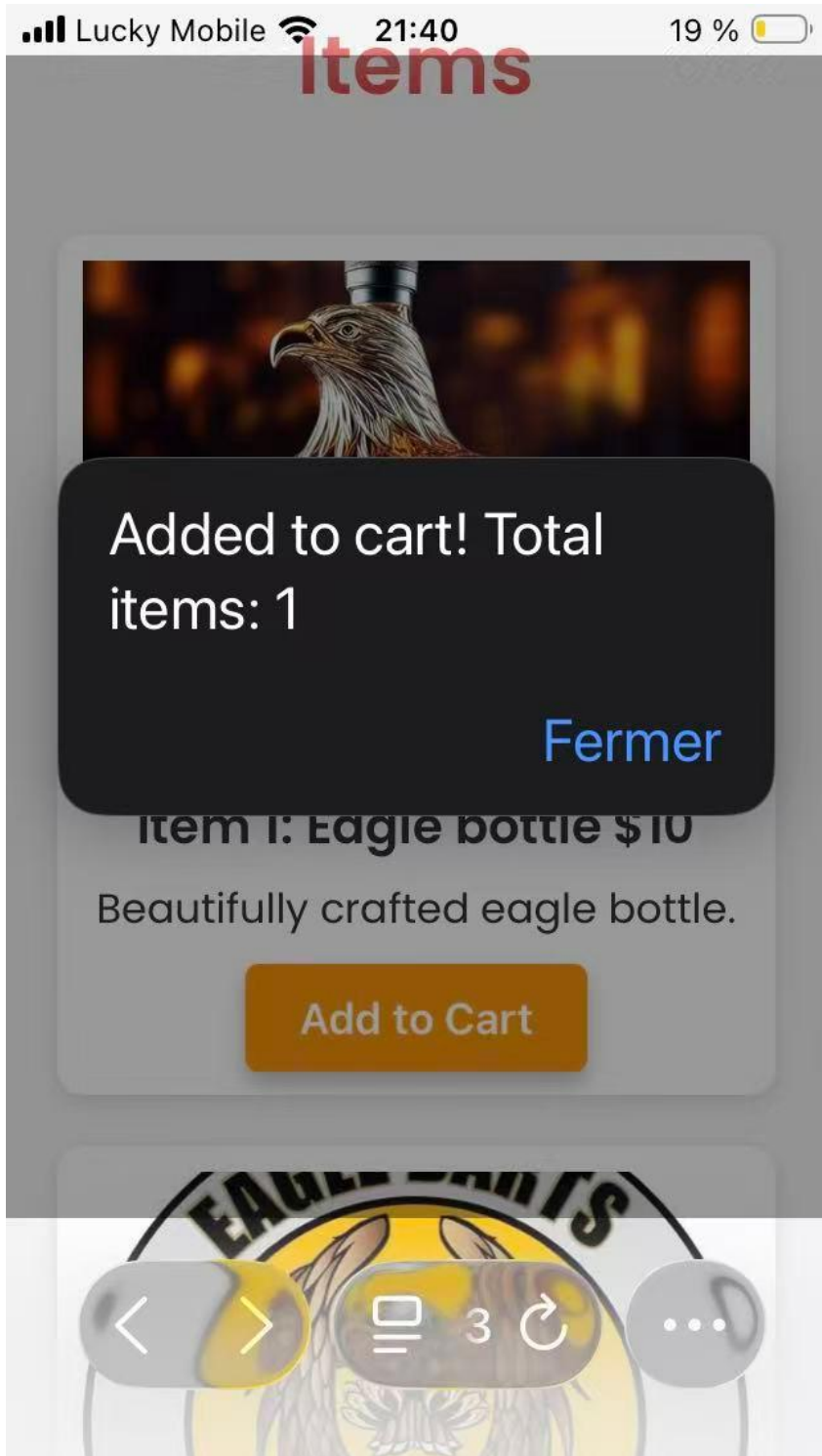
Beautifully crafted eagle bottle.

Add to Cart

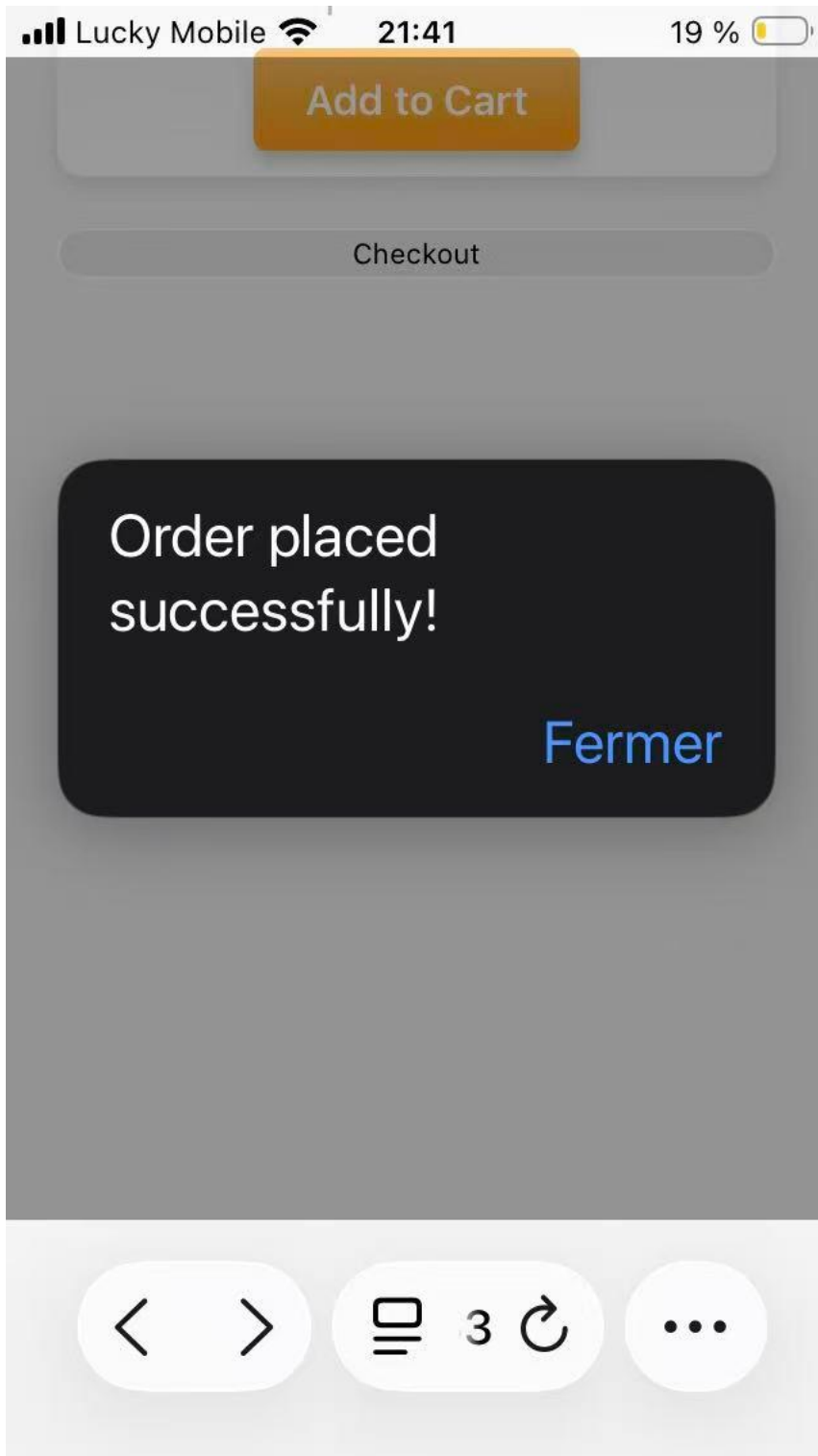


3





(Translation: Fermer means Close in French.)



(Translation: Fermer means Close in French.)

## Order History

[Refresh Orders](#)

Orders refreshed ✓

### Order #38

Date: 2026-06-16 01:40:59

[View Items](#) [Delete Order](#)

### Order #37

Date: 2026-06-16 01:39:40

[View Items](#) [Delete Order](#)

### Order #36

Date: 2026-06-16 01:39:14

[View Items](#) [Delete Order](#)

### Order #35

Date: 2026-06-16 01:33:39

[View Items](#) [Delete Order](#)

### Order #34

Date: 2026-06-16 01:28:42

[View Items](#) [Delete Order](#)

### Order #33

Date: 2026-06-16 01:15:35

[View Items](#) [Delete Order](#)

### Order #32

Date: 2026-06-16 01:14:46

[View Items](#) [Delete Order](#)

### Order #24

Date: 2026-04-19 19:18:21

[View Items](#) [Delete Order](#)

### Order #23

Date: 2026-04-19 19:17:06

[View Items](#)

48



3.0.53



### Order #22

Date: 2026-04-19 19:17:06

Order History

 Refresh Orders

Items loaded 

Order #38

Date: 2026-06-16 01:40:59

[View Items](#) [Delete Order](#)

- Eagle bottle - \$10 × 1 = \$10

Total: \$10

Order #37

Date: 2026-06-16 01:39:40

[View Items](#) [Delete Order](#)

Order #36

Date: 2026-06-16 01:39:14

[View Items](#) [Delete Order](#)

Order #35

Date: 2026-06-16 01:33:39

[View Items](#) [Delete Order](#)

Order #34

Date: 2026-06-16 01:28:42

[View Items](#) [Delete Order](#)

Order #33

Date: 2026-06-16 01:15:35

[View Items](#) [Delete Order](#)

Order #32

Date: 2026-06-16 01:14:46

[View Items](#) [Delete Order](#)

Order #24

Date: 2026-04-19 19:18:21

[View Items](#) [Delete Order](#)

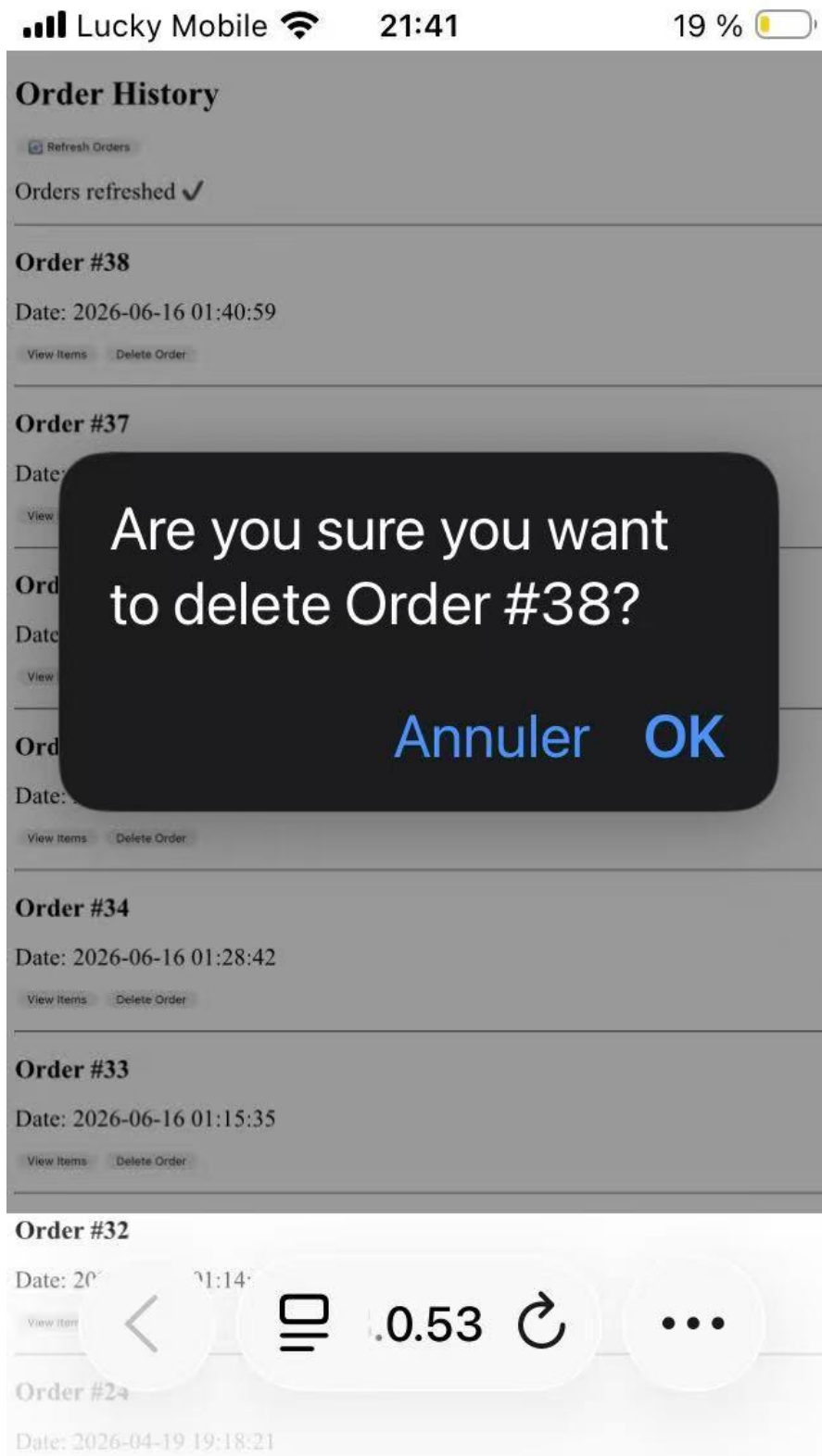
Order #23

Date: 2026-04-19 19:17:48

[View Items](#) [Delete Order](#)

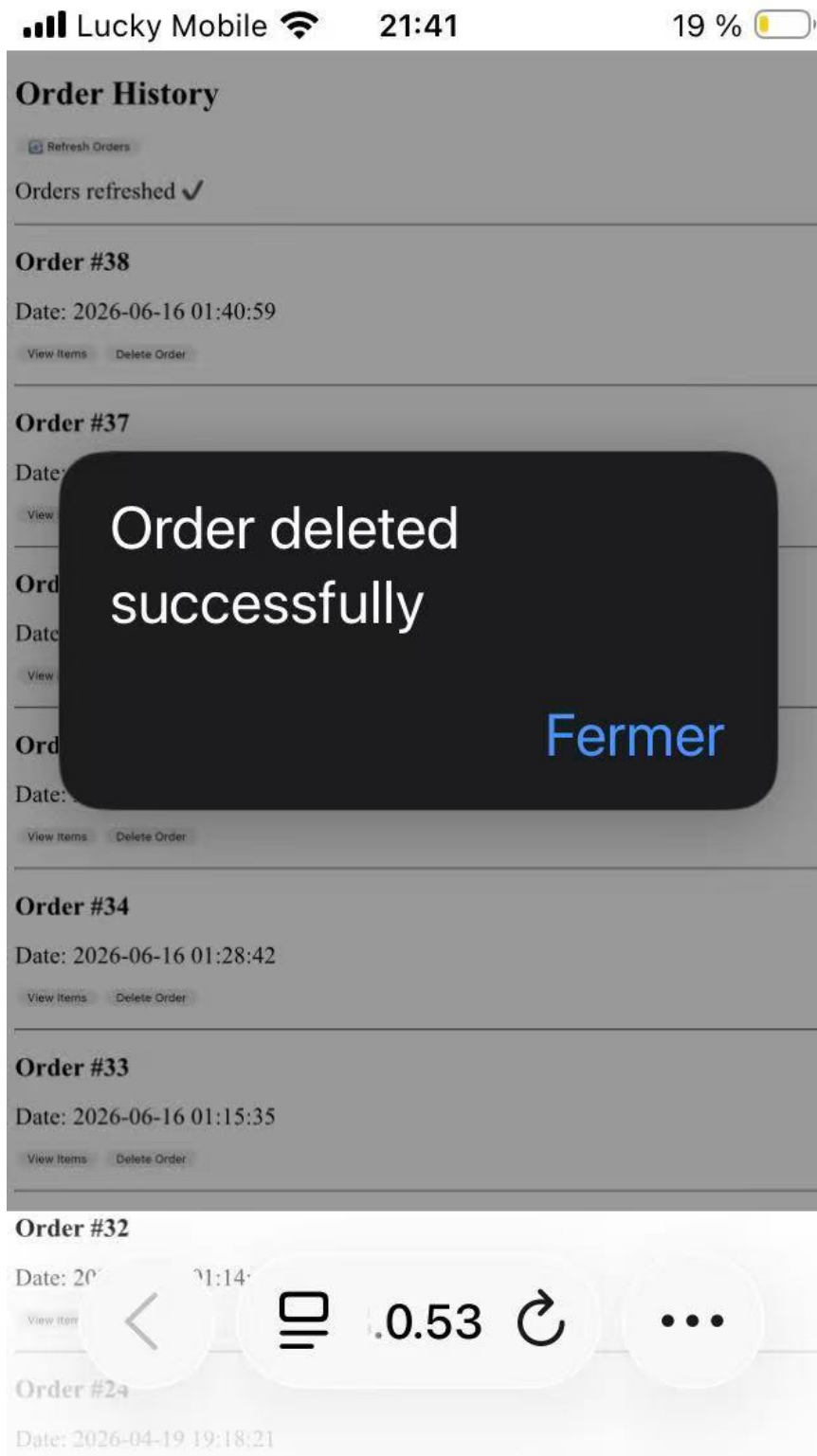
Order #22

192.168.0.53



(Translation: Annuler means Cancel in French.)





(Translation: Fermer means Close in French.)



## Order History

[Refresh Orders](#)

Orders refreshed ✓

### Order #37

Date: 2026-06-16 01:39:40

[View Items](#) [Delete Order](#)

### Order #36

Date: 2026-06-16 01:39:14

[View Items](#) [Delete Order](#)

### Order #35

Date: 2026-06-16 01:33:39

[View Items](#) [Delete Order](#)

### Order #34

Date: 2026-06-16 01:28:42

[View Items](#) [Delete Order](#)

### Order #33

Date: 2026-06-16 01:15:35

[View Items](#) [Delete Order](#)

### Order #32

Date: 2026-06-16 01:14:46

[View Items](#) [Delete Order](#)

### Order #24

Date: 2026-06-16 01:14:46

[View Item](#)

.0.53



### Order #23

Date: 2026-04-19 19:17:48

#### Localhost's reaction while running the application on the above browsers and iPhone:

1

```
C:\Users\lhuan\Desktop\446\20260414>node server.js
Server running on http://192.168.0.53:3000
Connected to SQLite database.
→ GET /
→ GET /external.css
→ GET /images/5eaglering.jpg
→ GET /images/1eaglebottle.jpg
→ GET /images/6eaglemug.jpg
→ GET /images/4eagleties.jpg
→ GET /jsfile.js
→ GET /images/3eagletshirt.jpg
→ GET /images/2eagledart.jpg
→ POST /cart
🛒 Add to cart: 1
→ POST /checkout
🛒 Checkout started
→ GET /orders.html
→ GET /orders?nocache=1781573324374
→ GET /orders/34?nocache=1781573328335
→ GET /
→ GET /external.css
→ GET /images/1eaglebottle.jpg
→ GET /jsfile.js
→ GET /images/2eagledart.jpg
→ GET /images/3eagletshirt.jpg
→ GET /images/4eagleties.jpg
→ GET /images/5eaglering.jpg
→ GET /images/6eaglemug.jpg
→ GET /favicon.ico
→ POST /cart
🛒 Add to cart: 1
→ POST /checkout
🛒 Checkout started
→ GET /orders.html
→ GET /orders?nocache=1781573624152
```

2

```
🛒 Add to cart: 1
➡ POST /checkout
🛒 Checkout started
➡ GET /orders.html
➡ GET /orders?nocache=1781573624152
➡ GET /orders/35?nocache=1781573629088
➡ POST /cart
🛒 Add to cart: 6
➡ POST /checkout
🛒 Checkout started
➡ GET /orders.html
➡ GET /orders?nocache=1781573955409
➡ GET /orders/36?nocache=1781573960619
➡ POST /cart
🛒 Add to cart: 6
➡ POST /checkout
🛒 Checkout started
➡ GET /orders.html
➡ GET /orders?nocache=1781573981975
➡ GET /orders/37?nocache=1781573994707
➡ POST /cart
🛒 Add to cart: 1
➡ POST /checkout
🛒 Checkout started
➡ GET /orders.html
➡ GET /orders?nocache=1781574064161
➡ GET /orders/38?nocache=1781574075986
➡ GET /orders.html
➡ GET /apple-touch-icon-precomposed.png
➡ GET /apple-touch-icon.png
➡ GET /favicon.ico
➡ GET /orders.html
➡ GET /apple-touch-icon-120x120-precomposed.png
➡ GET /apple-touch-icon-120x120.png
➡ GET /apple-touch-icon-precomposed.png
```

```
→ GET /apple-touch-icon.png
→ GET /favicon.ico
→ GET /apple-touch-icon-120x120-precomposed.png
→ GET /apple-touch-icon-120x120.png
→ GET /apple-touch-icon-precomposed.png
→ GET /apple-touch-icon.png
→ GET /favicon.ico
→ GET /orders.html
→ GET /orders?nocache=1781574096393
→ GET /apple-touch-icon-120x120-precomposed.png
→ GET /apple-touch-icon-precomposed.png
→ GET /apple-touch-icon.png
→ GET /favicon.ico
→ GET /apple-touch-icon-120x120.png
→ GET /apple-touch-icon-precomposed.png
→ GET /apple-touch-icon.png
→ GET /favicon.ico
→ DELETE /orders/38
Deleting order: 38
→ GET /orders?nocache=1781574117409
→ GET /
→ GET /external.css
→ GET /images/1eaglebottle.jpg
→ GET /images/2eagledart.jpg
→ GET /images/3eagletshirt.jpg
→ GET /images/4eagleties.jpg
→ GET /images/5eaglering.jpg
→ GET /jsfile.js
→ GET /images/6eaglemug.jpg
→ GET /undefined
→ GET /favicon.ico
→ POST /cart
🛒 Add to cart: 1
→ POST /checkout
```

```
→ GET /undefined
→ GET /favicon.ico
→ POST /cart
🛒 Add to cart: 1
→ POST /checkout
🛒 Checkout started
→ GET /orders.html
→ GET /orders?nocache=1781574222536
→ GET /undefined
→ GET /orders/39?nocache=1781574224499
→ DELETE /orders/39
Deleting order: 39
→ GET /orders?nocache=1781574231435
→ GET /
→ GET /external.css
→ GET /images/1eaglebottle.jpg
→ GET /images/2eagledart.jpg
→ GET /images/3eagletshirt.jpg
→ GET /images/4eagleties.jpg
→ GET /images/5eaglering.jpg
→ GET /jsfile.js
→ GET /images/6eaglemug.jpg
→ GET /undefined
→ POST /cart
🛒 Add to cart: 1
→ POST /checkout
🛒 Checkout started
→ GET /orders.html
→ GET /undefined
→ GET /orders?nocache=1781574858288
→ GET /orders/40?nocache=1781574862431
→ DELETE /orders/40
Deleting order: 40
→ GET /orders?nocache=1781574925067
```



```
→ GET /
→ GET /external.css
→ GET /images/1eaglebottle.jpg
→ GET /images/2eagledart.jpg
→ GET /images/3eagletshirt.jpg
→ GET /images/5eaglering.jpg
→ GET /images/4eagleties.jpg
→ GET /jsfile.js
→ GET /images/6eaglemug.jpg
→ POST /cart
🛒 Add to cart: 1
→ POST /checkout
🛒 Checkout started
→ GET /orders.html
→ GET /orders?nocache=1781576909737
→ GET /
→ GET /external.css
→ GET /jsfile.js
→ GET /images/1eaglebottle.jpg
→ GET /images/6eaglemug.jpg
→ GET /images/5eaglering.jpg
→ GET /images/4eagleties.jpg
→ GET /images/3eagletshirt.jpg
→ GET /images/2eagledart.jpg
→ GET /favicon.ico
→ POST /cart
🛒 Add to cart: 1
→ POST /checkout
🛒 Checkout started
→ GET /orders.html
→ GET /orders?nocache=1781577093473
→ GET /orders/42?nocache=1781577112516
→ DELETE /orders/42
Deleting order: 42
→ GET /orders?nocache=1781577159623
```

### ### Debugging issue one:

Buttons on iPhone are not reacting.

#### #### Solution:

Change all the localhost address in the application to the laptop local IPv4 address, which is 192.168.0.53:3000.

### ### Debugging issue two:

The dates show invalid date.

#### #### *Solution:*

removed " UTC"

changed the date to,

```
<p>Date: ${new Date(order.created_at).toLocaleString()}</p>
```

---

## ## Ask Others to Test My Application

I shared my application with my parents, and it ran well on their iPhone Safaris too.

## ## Future Improvements

- Improve mobile responsiveness for product listing, cart, and checkout pages
- Optimize performance by reducing unnecessary database queries and improving load times
- Add analytics features to track user activity such as most viewed products and cart conversions

---

## ## GitHub link and live link

Repository Links:



frontend:

[https://github.com/labricot-lab/m5\\_lab5\\_20260310.git](https://github.com/labricot-lab/m5_lab5_20260310.git)

[https://github.com/labricot-lab/446\\_m5\\_coursepro\\_20260312.git](https://github.com/labricot-lab/446_m5_coursepro_20260312.git)

backend:

[https://github.com/labricot-lab/446\\_pro\\_08\\_20260421.git](https://github.com/labricot-lab/446_pro_08_20260421.git)

[https://github.com/labricot-lab/446\\_pro\\_07\\_20260419.git](https://github.com/labricot-lab/446_pro_07_20260419.git)

[https://github.com/labricot-lab/446\\_pro8\\_updated\\_20260506.git](https://github.com/labricot-lab/446_pro8_updated_20260506.git)

Repository live links:

The application was tested locally using:

<http://localhost:3000>

The application was also tested globally using:

<http://192.168.0.53:3000>

frontend:

[https://labricot-lab.github.io/m5\\_lab5\\_20260310/](https://labricot-lab.github.io/m5_lab5_20260310/)

[https://labricot-lab.github.io/446\\_m5\\_coursepro\\_20260312/](https://labricot-lab.github.io/446_m5_coursepro_20260312/)

backend:

[https://labricot-lab.github.io/446\\_pro\\_08\\_20260421/](https://labricot-lab.github.io/446_pro_08_20260421/)

[https://labricot-lab.github.io/446\\_pro\\_07\\_20260419/](https://labricot-lab.github.io/446_pro_07_20260419/)

[https://labricot-lab.github.io/446\\_pro8\\_updated\\_20260506/](https://labricot-lab.github.io/446_pro8_updated_20260506/)

the whole project:

[https://github.com/labricot-lab/446pro9\\_20260615](https://github.com/labricot-lab/446pro9_20260615)